

Sprint 1 Readiness

Project: Image Guided Surgery Intervention

Candidate Stories

Story ID	Title	SP	Acceptance Criteria	Dependencies	Risks and Mitigations
S1-01	Familiarization with Existing Software	8	Given access to existing software repository and documentation, When the modules and data flow will be examined, Then the high-level architecture can be documented and its core components identified.	Access to existing code repository and documentation.	Complicated code -> Pair review sessions
S1-02	Setup and Documentation	5	Given a fully functional, controlled environment with Python and other dependencies installed, When the system runs without error, Then the setup process will be documented so the environment can be replicated. Given the documented setup process and fully functional environment, When the code executes as expected, Then further documentation can be written about code execution	Environment configuration files.	Dependency mismatch -> Docker or Virtual Environment Implementation
S1-03	Modularization and Extensibility	13	Given the current code architecture, When core components are refactored into separate modules, Then each module can be imported and tested independently.	Refactoring plan.	Integration issues -> Incremental refactor

			Given the refactored structure, When new functionality is added, Then it can be easily integrated with existing modules without modifying them.		
S1-04	Initial Database Setup	5	Given the need for a scan storage, When the database schema is designed and initialized locally, Then the application can successfully connect to the DB instance.	Story S1-02 (Setup)	Connection latency or storage limitation -> Use SQLite for local dev.
S1-05	Unit Testing Framework	5	Given the refactored and restructured project environment, When unit tests are created and run against each module, Then all core functions achieve high pass rates. Given all integration tests executed, When the project pipeline is tested in a simulated environment, Then the output matches expected results with no runtime errors.	Story S1-03	Low test coverage -> Frequent coverage reviews

KPI /OKR Baseline

Objective	Key Result (Baseline -> Target)	Measure	Owner
Establish Development Environment	0 -> 100% reproducible setup by end of Sprint 1	Successful setup on all group member's machines	Dev Lead
Improve Code Modularity	Monolithic -> Distinct individual modules	Module build status	Tech Lead

Ensure Software Reliability	0 -> 80% Test coverage by end of Sprint 1	Test coverage report	QA Lead
Database Foundation	No Storage -> Connected Schema	Successful connection and scan storage	DB Lead